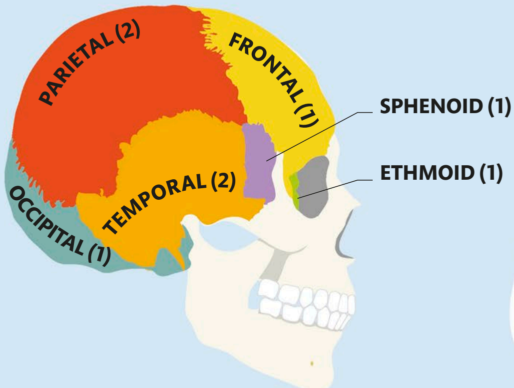
Protecting the Brain

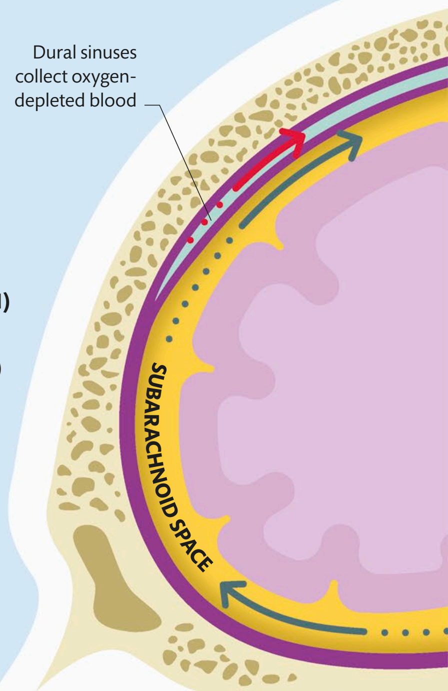
The vital organs are safely secured in the body's core, but because the brain sits in the head at the top of the body, it requires its own protection system.

The cranium

The bones of the head are collectively known as the skull but are more correctly divided into the cranium and the mandible, or jawbone. It is supported by the highest cervical vertebra and the musculature of the neck. The cranium forms a bony case completely surrounding the brain. It is made of 22 bones that steadily fuse together in the early years of life to make a single, rigid structure. Nevertheless, the cranium has around 64 holes, known as foramina, through which nerves and blood vessels pass, and eight air-filled voids, or sinuses, which reduce the weight of the skull.



Paired bones
The brain is enclosed by eight large bones, with a pair of parietal and temporal bones forming each side of the cranium. The remaining 14 cranial bones make up the facial skeleton.



2 Direction of flow
CSF flows from the ventricles into the subarachnoid space, where it then moves up and over the front of the brain.

Cerebrospinal fluid

The brain does not come into direct contact with the cranium. Instead it is suspended in cerebrospinal fluid (CSF). This clear liquid circulating inside the cranium creates a cushion around the brain to protect it during impacts to the head. In addition, the floating brain does not deform under its own weight, which would otherwise restrict blood flow to the lower internal regions. The exact quantity of CSF also varies to maintain optimal pressure inside the cranium. Reducing the volume of CSF lowers the pressure, which in turn increases the ease with which blood moves through the brain.

WHAT IS WATER ON THE BRAIN?

Also called hydrocephalus, this condition arises when there is too much CSF in the cranium. This puts pressure on the brain and affects its function.

Meninges and ventricles
The brain is surrounded by three membranes, or meninges: the pia mater, arachnoid mater, and dura mater. The CSF fills cavities called ventricles and circulates around the outside of the brain in the subarachnoid space, which lies between the pia and arachnoid mater.

CSF IS CONTINUALLY PRODUCED, AND ALL OF IT IS REPLACED EVERY 6–8 HOURS